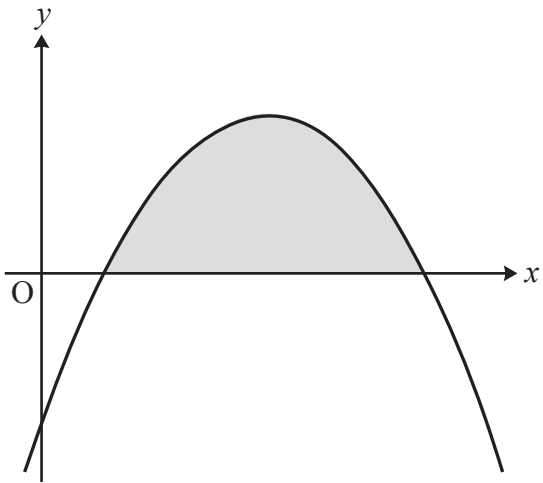
7 In this question you must show detailed reasoning.



Find the exact area of the shaded region shown in the diagram, enclosed by the x-axis and the curve $y = -3x^2 + 7x - 2$. **[6]**